Homework 2

October 10, 2026

Some of the problems are collected from the examples in the book.

Problem 1

Let $\mathscr{A} \in \mathcal{L}(\mathcal{C}_2(t) \to \mathcal{C}_2(t))$. With respect to the bases $\alpha_1(t)$, $\alpha_2(t)$ and $\alpha_3(t)$ with

$$\alpha_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \alpha_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

the matrix representation is

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Determine $\mathcal{N}(\mathscr{A})$ and $\mathcal{R}(\mathscr{A})$.

Problem 2

Show the following propositions:

(1) $A \in \mathcal{F}^{n\times n}$, then $\mathcal{S} = \{\boldsymbol{x}(t) : \dot{\boldsymbol{x}}(t) = A\boldsymbol{x}(t)\}$ is a linear space and is isomorphic with $\mathcal{C}^n$, where

$$\dot{\boldsymbol{x}}(t) \triangleq \begin{bmatrix} \dfrac{dx_1(t)}{dt} & \dfrac{dx_2(t)}{dt} & \cdots & \dfrac{dx_n(t)}{dt} \end{bmatrix}^T.$$

(2) $A \in \mathcal{C}^{n\times n}$, $B \in \mathcal{C}^{m\times m}$ are given matrices, $\mathcal{T}_{A,B} : \mathcal{C}^{n\times m} \to \mathcal{C}^{n\times m}$ is defined as

$$\mathcal{T}_{A,B}(X) = AX + XB,$$

then $\mathcal{T} \in \mathcal{L}\big(\mathcal{C}^{n\times m} \to \mathcal{C}^{n\times m}\big)$.

Problem 3

Given the following two sets of bases for $\mathcal{V}^3$:

$$\epsilon_1 = (1,0,1)^T, \quad \eta_1 = (1,2,-1)^T$$

$$\epsilon_2 = (2,1,0)^T, \quad \eta_2 = (2,2,-1)^T$$

$$\epsilon_3 = (1,1,1)^T, \quad \eta_3 = (2,-1,-1)^T$$

define the linear transformation as:

$$\mathscr{A}\epsilon_i = \eta_i, \qquad (i = 1,2,3).$$

(1) Find the transition matrix from bases $\epsilon_1, \epsilon_2, \epsilon_3$ to bases η_1, η_2, η_3;

(2) Give the matrix representation for $\mathscr{A}$ with respect to the bases $\epsilon_1, \epsilon_2, \epsilon_3$;

(3) Give the matrix representation for $\mathscr{A}$ with respect to the bases η_1, η_2, η_3.

Problem 4

Denote by $\mathcal{R}_n[\lambda]$ the set of all polynomials with real coefficients and degrees not exceeding n. Let a linear transformation $\mathscr{A} \in \mathcal{L}\big(\mathcal{R}_n[\lambda] \mapsto \mathcal{R}_n[\lambda]\big)$ be defined as

$$\mathscr{A}(f(\lambda)) = \frac{d}{d\lambda} f(\lambda), \qquad f(\lambda) \in \mathcal{R}_n[\lambda],$$

where $\mathcal{L}(\cdot)$ is the set of linear transformations. When $n = 2$, compute

(1) The matrix representation of $\mathscr{A}$ with respect to the basis $\alpha_1(\lambda) = 1 + \lambda + \lambda^2$, $\alpha_2(\lambda) = 1 + \lambda$, $\alpha_3(\lambda) = 1 + 2\lambda$.

(2) Determine $\mathcal{N}(\mathscr{A})$ and $\mathcal{R}(\mathscr{A})$ of $\mathscr{A}$.

Problem 5

Let $\mathscr{A}$ be a linear transformation from a four-dimensional linear space $\mathcal{V}$ to itself, $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be a basis of the linear space $\mathcal{V}$, and the matrix representation of $\mathscr{A}$ with respect to this basis is

$$A = \begin{bmatrix} 5 & -2 & -4 & 3 \\ 3 & -1 & -3 & 2 \\ -3 & \frac{1}{2} & \frac{9}{2} & -\frac{5}{2} \\ -10 & 3 & 11 & -7 \end{bmatrix}$$

(1) Find the matrix representation of $\mathscr{A}$ with respect to the following basis

$$\eta_1 = \alpha_1 + 2\alpha_2 + \alpha_3 + \alpha_4$$

$$\eta_2 = 2\alpha_1 + 3\alpha_2 + \alpha_3$$

$$\eta_3 = \alpha_3$$

$$\eta_4 = \alpha_4$$

(2) Determine $\mathcal{N}(\mathscr{A})$ and $\mathcal{R}(\mathscr{A})$ of $\mathscr{A}$.